

data using REST API calls. Fully customisable, ThingsBoard clusters offer possibilities to create a range of technical microservices -- HTTP/MQTT/CoAP transport microservices, WebUI microservices and JavaScript executor microservices. Rule based data processing algorithms can be applied to normalise, validate or transform input data sets. Users can also customise the Rule Engine toolset from the ThingsBoard dashboard to drag/drop Rule Nodes or define Root Rule Chain.

ThingsBoard is best known for its real-time IoT dashboard. The toolkit offers more than thirty customisable widgets to create rich visualisations, perform deep analytics and provide compelling IoT use cases.

ThingSpeak

The data aggregation and analytics IoT toolkit, ThingSpeak, offers non-commercial open source solutions that can visualise IoT device data using MATLAB widgets. ThingSpeak's reputation can be attributed to its seamless integration with the MathWorks product suite. This robust IoT analytics tool supports RaspberryPi, Arduino and Nodemcu devices.

IoT sensor data transferred to the ThingSpeak cloud using restful APIs and HTTP protocols can be analysed and visualised for more in-depth insights using MATLAB software. There are also options to retrieve data in JSON, XML and CSV formats for manual data analysis and reporting. There are options for users to share data with their teams using private and public channels. ThingSpeak also has a paid commercial toolkit, but its open source and free-to-use solutions that work alongside MATLAB computational algorithms are more than well-suited for performing the fundamental IoT data analysis and visualisations.

Apache StreamPipes

This industrial analytics toolkit is known to help both non-technical and technical users to collect, analyse and study IoT data

sets. StreamPipes uses machine learning algorithms to perform advanced analytics, pattern detection, predictive analysis, anomaly detection and temporal analysis. It is well-reputed with non-technical users thanks to the intuitive, easy to use Web interface and graphical editor.

StreamPipes Connect, the in-built channelisation framework, can collect data inputs both from IoT device archived and real-time data sets. StreamPipes also comes with built-in semantics to provide intelligent insights and recommendations for data stream elements and/or transformational modules. The toolkit is compatible with HTTP/REST, MQTT, Kafka, OPCUA and ROS protocols. Enterprise key process indicators (KPIs) and production reports can be visualised in real-time using Web based cockpits.

There are additional options for software developers to use wrappers like Apache Flink and Apache Spark to customise SDKs and Maven archetypes to create new data processing elements. One of the unique features of StreamPipes is its ability to aggregate geographically distributed data pipelines in real-time, thereby creating possibilities to perform edge computing on IoT data. Various data harmonisation algorithms like filters, aggregation and unit converters, help developers clean and enrich device sensor data, periodically.

WSO2 IoT Server

A server for the IoT platform released under Apache 2.0 license, this toolkit is trusted to offer versatile solutions with edge computing. WSO2 IoT Server creators pride themselves on its seamless integration, easy-to-deploy drag/drop widgets and platform scalability. The platform can manage up to a million IoT devices and provide deep data analytics of all the data aggregated from them.

WSO2 uses WSO2 Data Analytics Server (WSO2 DAS) to perform real-time analysis, batch analysis, interactive analysis and predictive analytics. WSO2 Complex Event Processor (WSO2 CEP) is used to handle millions of data aggregations per second. This makes the analytics platform well-suited to processing enormous volumes of IoT data.

WSO2 also provides analytics extension event adapters for HBase, Rabbitmq and Twitter, in addition to the native built-in event adapters available on its analytics platform.

Previously, the processes of collecting, storing and analysing an enormous volume of data sets was considered a complex and expensive task. But, today, with IoT standardisation, cloud computing, machine learning and edge computing, IoT analytics is taking huge progressive leaps in the industry. Our commercial world may not be fully adapted yet to leverage the power of IoT analytics, but we are definitely getting there.

Industries such as retail, pharma, healthcare, manufacturing and even smart city projects are increasingly gaining momentum in terms of artificial intelligence, machine learning and data analytics. Fortune 500 companies are already deploying IoT architecture in order to have a better understanding of their business processes and customer preferences. IoT data analytics is transforming enterprises and businesses already. Therefore, the sooner we adapt to achieving new possibilities with IoT, the better our stakes will be to make early gains from them.

What would be a better way to go SMART with IoT than by trying it for free? Try your hands on these top five open source IoT analytics tools and upgrade yourself for the future of work. **END** 🐧

 By: Beulah Evelyn

The author is a research scholar, writer and geo-spatial technology fanatic.